



Effects of exposure to malathion on blood glucose concentration: a meta-analysis

Marco Antonio Ramirez-Vargas¹ · Eugenia Flores-Alfaro² · Mayrut Uriostegui-Acosta³ · Patricia Alvarez-Fitz¹ · Isela Parra-Rojas⁴ · Ma. Elena Moreno-Godínez¹ 

Received: 16 May 2017 / Accepted: 29 November 2017 / Published online: 12 December 2017
© Springer-Verlag GmbH Germany, part of Springer Nature 2017

Abstract

Exposure to malathion (an organophosphate pesticide widely used around the world) has been associated with alterations in blood glucose concentration in animal models. However, the results are inconsistent. The aim of this meta-analysis was to evaluate whether malathion exposure can disturb the concentrations of blood glucose in exposed rats. We performed a literature search of online databases including PubMed, EBSCO, and Google Scholar and reviewed original articles that analyzed the relation between malathion exposure and glucose levels in animal models. The selection of articles was based on inclusion and exclusion criteria. The database search identified thirty-five possible articles, but only eight fulfilled our inclusion criteria, and these studies were included in the meta-analysis. The effect of malathion on blood glucose concentration showed a non-monotonic dose-response curve. In addition, pooled analysis showed that blood glucose concentrations were 3.3-fold higher in exposed rats than in the control group (95% CI, 2–5; $Z = 3.9$; $p < 0.0001$) in a random-effect model. This result suggested that alteration of glucose homeostasis is a possible mechanism of toxicity associated with exposure to malathion.

Keywords Malathion · Organophosphate pesticides · Hyperglycemia · Animal models

Introduction

Disruption of energetic metabolism has been considered a mechanism through which organophosphate pesticides (OP) exert their toxicological effects (Romero-Navarro et al. 2006;

Basiri et al. 2007; Karami-Mohajeri and Abdollahi 2011; Lasram et al. 2015b; Mostafalou and Abdollahi 2016). Malathion (diethyl (dimethoxyphosphinothioylthio) succinate) is a broad-spectrum OP classified by the World Health Organization as a slightly hazardous pesticide, and this OP is used widely in agriculture and brigades to control vector-borne diseases (WHO 2010). The neurotoxic effects induced by exposure to malathion are caused by the inhibition of acetylcholinesterase in the central and peripheral nervous systems (Karami-Mohajeri and Abdollahi 2011; Karami-Mohajeri et al. 2014; Fakhri-Bafghi et al. 2016).

In addition, acute, subchronic, or chronic exposure to malathion has been associated with metabolic perturbations in exposed populations (Namba 1971; Raafat et al. 2012) and animal models (Gupta 1974; Rezg et al. 2007; Lasram et al. 2015a); the most frequently found alteration is hyperglycemia (Abdollahi et al. 2004; Mostafalou et al. 2012; Lasram et al. 2015b). Hyperglycemia is a pathophysiologic state characterized by an abnormal increase in blood glucose concentrations; this state has a key role in the development of chronic diseases, such as diabetes mellitus type 2, stroke, and cardiovascular diseases (Levitan et al. 2004; Barclay et al. 2008; Stead et al. 2009).

Responsible editor: Philippe Garrigues

✉ Ma. Elena Moreno-Godínez
emoreno20@hotmail.com

¹ Laboratorio De Toxicología y Salud Ambiental, Facultad De Ciencias Químico Biológicas, Universidad Autónoma De Guerrero, Av. Lázaro Cárdenas s/n, Ciudad Universitaria, 39070 Chilpancingo, Guerrero, Mexico

² Laboratorio De Investigación En Epidemiología Clínica y Molecular, Facultad De Ciencias Químico Biológicas, Universidad Autónoma De Guerrero, Chilpancingo, Mexico

³ Laboratorio de Inmunotoxicogenómica, Escuela Superior de Ciencias Naturales, Universidad Autónoma De Guerrero, Chilpancingo, Mexico

⁴ Laboratorio De Investigación En Obesidad y Diabetes, Facultad De Ciencias Químico Biológicas, Universidad Autónoma De Guerrero, Chilpancingo, Mexico

Peripheral tissue and hepatic insulin resistance or enhanced hepatic glucose production have been associated with the induction of hyperglycemia states (Brealey and Singer 2009). Insulin resistance is associated with multiple risk factors, such as genetic factors, a sedentary lifestyle, a Western diet, and obesity (Petersen and Shulman 2006), while enhanced hepatic glucose production occurs in response to hepatic or tissue damage (Li and Messina 2009).

Exposure to malathion in animal models has proven to be a risk factor for the development of hyperglycemia by inducing insulin resistance (Mostafalou et al. 2012; Lasram et al. 2015a) and enhancing hepatic glucose production (Abdollahi et al. 2004; Pournourmohammadi et al. 2005; Vosough-Ghanbari et al. 2007), although results have been inconsistent (Rezg et al. 2006, 2007; Flehi-Slim et al. 2015). Thus, it is necessary to perform a systematic review and meta-analysis of the results obtained in animal models to provide an external validation of primary studies, which could reduce the risk of publication bias (Hooijmans et al. 2014a). In the present work, we performed a systematic review and meta-analysis to arrive at pooled estimates for the effect of malathion exposure on blood glucose concentrations in animal models.

Materials and methods

To carry out this work, we followed the recommendations of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Liberati et al. 2009). A literature search was performed on databases (PUBMED, EBSCO, Google Scholar) (up to 12/2016) using the following term restrictions: “organophosphates pesticides,” “malathion,” “glucose,” “impaired glucose toleration,” “blood glucose,” “insulin resistance,” and “animal models.” The terms were grouped into three principal components “malathion,” “blood glucose concentration,” and “animal models.” We used the Medical Subject Headings (MeSH) terms; in addition, the most frequent synonyms were added. The full strategy for each individual component is presented in the Table 1. The search also included the review of the cited bibliographies to identify omitted articles.

Selection criteria and data extraction

The selection criteria were as follows: original articles that (1) administered malathion to rats, (2) provided the number of animals used for each experimental group, (3) and reported the administered dose and time of exposure. When duplicate works of a research group existed, only the most recent report was considered. The result of interest was the blood glucose concentration expressed as the mean and standard error (or the data for their estimation). Data were screened and extracted

independently by two investigators to identify and determine if a study was included in this meta-analysis. Differences in data extraction between investigators were infrequent and were resolved by consensus. The following data was extracted from each study: the publication year, name of the first author, number of animals for each experimental group, dose administered, time of exposure, and blood glucose concentration.

Methodological quality assessment

Two independent reviewers submitted the studies that fulfilled our inclusion criteria to quality assessment. We used the “SYRCLE’s risk of bias tool for animal studies” (Hooijmans et al. 2014b), as a standardized method for quantification of the risk of bias. The quality assessment for all the included studies was expressed as a percentage. The division of the number of articles that fulfilled with an item described in SYRCLE’s risk tool and the total number of papers included was used for estimated this percentage.

Statistical analysis

We performed a meta-regression with random effects to estimate the changes in blood glucose concentrations from exposure to malathion. The statistical model was $y_i = \mu + \beta x_i + \tau_i + \varepsilon_i$; this model allows for the estimation of the variability between studies, and proportioned estimations of the parameters were considered (Harbord and Higgins 2008). The mean and linear splines were used to assess the dose-response relationship; because this model does not have an “a priori” statistical function, the x-axis was divided into equal-width intervals and then the median of y and the median of x were estimated in each interval (Gould 1993). In addition, the mean differences and their 95% confidence intervals (95% CIs) of blood glucose concentration were estimated in each study using random effects; for this estimation, the I^2 test was performed to measure heterogeneity, as this parameter indicates the proportion of variation between studies with respect to total variation (Sterne et al. 2008). Statistical analyses were performed using the statistical package STATA version 13.0 (College Station, TX).

Results

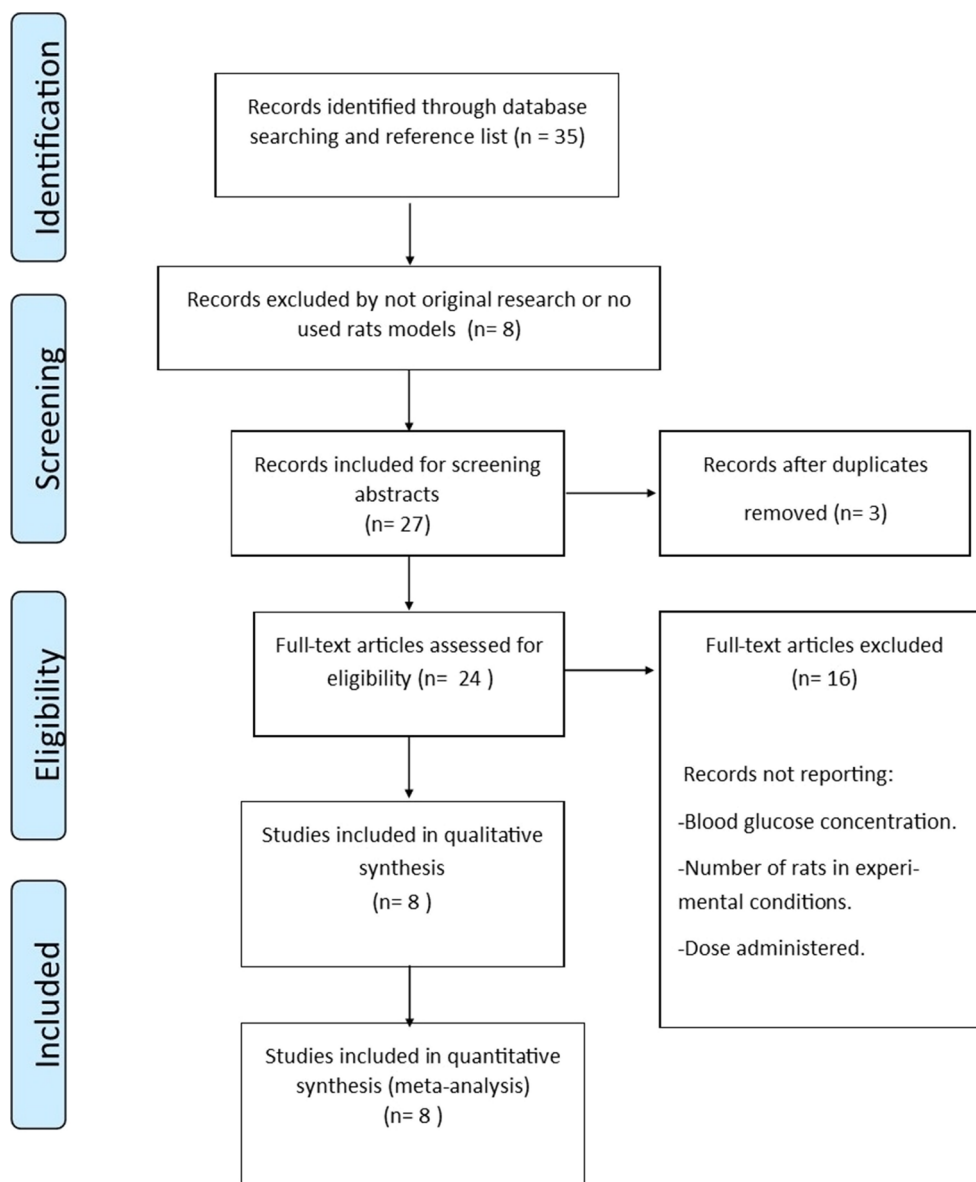
We identified 35 relevant articles through searching databases, and eight studies fulfilled our inclusion criteria and were included in the meta-analysis (Fig. 1). All studies used rats (albino Wistar rats) as an experimental model. In addition, malathion was administered orally, intramuscularly or intraperitoneally, and the control groups received only the vehicle control without malathion. Each study reported the measurement of blood glucose concentrations after exposure to malathion.

Table 1 Terms of restriction used for screening in the database

Component 1: “Malathion”	“malathion”[MeSH Terms] OR “malathion”[All Fields]
Component 2: “Blood glucose concentration”	((“glucose”[MeSH Terms] OR “glucose”[All Fields]) OR (“blood glucose”[MeSH Terms] OR (“blood”[All Fields] AND “glucose”[All Fields]) OR “blood glucose”[All Fields]) OR (“insulin resistance”[MeSH Terms] OR (“insulin”[All Fields] AND “resistance”[All Fields]) OR “insulin resistance”[All Fields]))
Component 3: “Animal models”	(“rats”[MeSH Terms] OR “rats”[All Fields] OR “rat”[All Fields]) AND “animals”[MeSH Terms:noexp]

The assessment of methodological quality of the studies included in this review showed that these have a low risk of bias (Fig. 2). These works meet most of the criteria for assessing the methodological quality according to the SYRCLÉ’s risk of bias tool for animal studies. However, the items “Was the allocation to the different groups adequately concealed

during?, Were the caregivers and/or investigators blinded from knowledge which intervention each animal received during the experiment?, Were animals selected at random for outcome assessment?” showed a 100% of the unclear risk of bias.

Fig. 1 Flow diagram of the selection and eligibility of studies

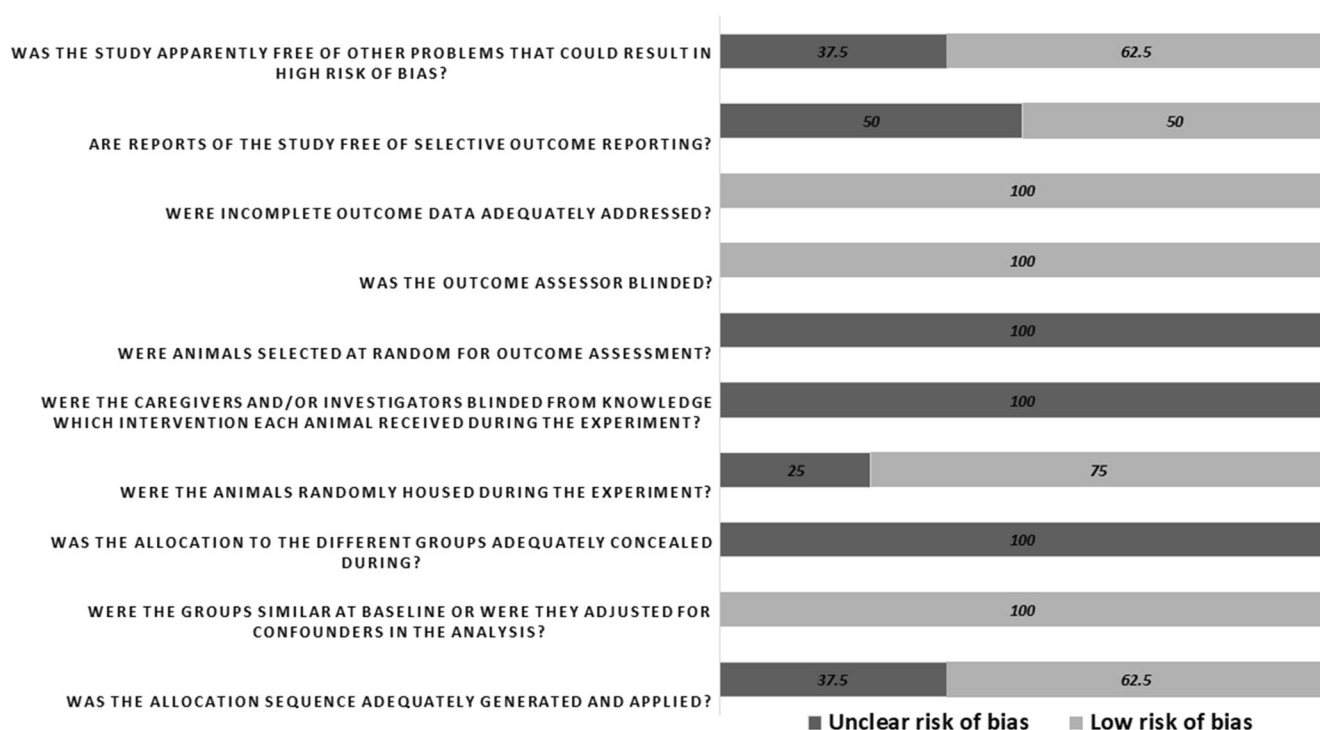


Fig. 2 Methodological quality of included studies using SYRCLE's risk of bias tool. The risk of bias as a percentage for each item contemplate is shown

The time of exposure varied among the studies and was categorized as acute (<48 h) or chronic (>48 h) exposure; the dose administered was also diverse and was categorized and reported for rats as low dose (≤ 200 mg/kg) or high dose (> 200 mg/kg), considering a 1/10 of the lethal dose 50 (2000 mg/kg, reviewed in WHO, 2009), Table 2. Alteration in some metabolic parameters in animal models exposed to malathion is shown (Table 3).

As shown in Fig. 3, a non-monotonic dose-response curve comparing blood glucose concentrations and malathion dose was determined. At low doses, a semi-linear response was observed. However, at higher doses of exposure, the blood glucose concentrations decreased. The results of meta-regression showed a significant effect of exposure to malathion on blood glucose concentration ($\beta = 0.21$; 95% CI, 0.1–0.3, $p = 0.001$; $R^2_{\text{adj}} = 40.42\%$).

Stratified analyses of the time of exposure showed that chronic exposure of malathion increased the effect on blood glucose concentration ($\beta = 0.4$, 95% CI, 0.06–0.7, $p = 0.025$; $R^2_{\text{adj}} = 23.03\%$) compared to that of acute exposure ($\beta = 0.2$, 95% CI, 0.2–0.3, $p = 0.004$; $R^2_{\text{adj}} = 98.98\%$).

A pooled analysis using a random effects model showed that blood glucose concentrations were 3.3-fold higher in exposed rats than in the control group (95% CI, 2–5; $Z = 3.86$; $p < 0.0001$), and significant heterogeneity was observed ($p < 0.0001$, $I^2 = 89.6\%$) (Fig. 4).

Discussion

The present study focused on determining the abnormal increase of blood glucose concentrations after exposure to malathion in animal models. Hyperglycemia is an abnormal state of an increase in blood glucose concentrations. Long and frequent periods of hyperglycemia are associated with the development of chronic diseases (Levitan et al. 2004; Barclay et al. 2008; Stead et al. 2009), and the pathogenic mechanism is by the induction of insulin resistance (Tomás et al. 2002).

The effect of malathion on blood glucose concentration followed a non-monotonic dose-response curve. Exposure to doses less than 200 mg/kg showed a semi-linear dose-response effect. An increase in blood glucose concentration could be attributed to enhanced hepatic glucose production by the overstimulation of enzymes implicated in glycogenolysis (such as glucogen phosphorylase) or in neoglucogenesis (such as phosphofructokinase) in the liver (Abdollahi et al. 2004; Basiri et al. 2007) or muscle tissue (Pournourmohammadi et al. 2005) at low doses. However, exposure to high doses of malathion (> 200 mg/kg) could have decreased blood glucose concentrations by liver injury (Bissell et al. 2001). Along these lines, chickens exposed to 10 mg/kg malathion for 60 days showed liver necrosis (Sodhi et al. 2008). Additionally, hypoglycemia has been reported in chicken embryos exposed to malathion, and this reduction in blood glucose concentrations could be due to liver injury (Arsenault et al. 1975). A similar effect has been reported

Table 2 Characteristics of the studies contributing to the analysis of the effect of malathion exposure on blood glucose concentration

Study, year	Administered dose (mg/kg)	<i>n</i> ^a	Time of exposure (h)		Blood glucose concentration (mg/dL) ^b
			Acute (≤ 48 h)	Chronic (> 48 h)	
Basiri et al. (2007)	0	4		432	62.2 (3.11)
		4		672	63.4 (3.17)
	20	4		432	76.4 (3.8)
		4		672	72.2 (3.61)
Flehi-Slim et al. (2015)	0	6		720	162.14 (4.93)
	1.3	6		720	164.12 (1.91)
	13.7	6		720	176.91 (4.63)
	137	6		720	192.77 (7.94)
Lasram et al. (2015b)	0	12		672	107 (2)
	200	12		672	196 (5)
Matin and Siddiqui (1982)	0	6	0.5		95.1 (2.9)
	500	6	0.5		215 (10.9)
Matin and Husain (1987)	0	6	2		94.6 (4.06)
	500	6	2		200.5 (3.34)
Mostafalou et al. (2012)	0	6		768	89.16 (5.7)
	25	6		768	119.5 (2.61)
	50	6		768	112.3 (4.13)
Poumoumohammadi et al. (2005)	0	8		672	102.7 (5.5)
	10	8		672	148.3 (6.6)
	20	8		672	164.9 (7.7)
Rezg et al. (2007)	0	16		768	116 (24)
	100	16		768	120 (15)

^a Number of albino Wistar rats for each experimental group^b Value displayed as the mean and standard error

in animals exposed to other OP such as diazinon (Teimouri et al. 2006; Ghafour-Rashidi et al. 2007), chlorpyrifos (Acker and Nogueira 2012), monocrotophos (Joshi and Rajini 2012), acephate (Joshi and Rajini 2009), and parathion (Lassiter et al. 2008). The overstimulation of glycogenolysis enzymes was linked with the increase of blood glucose concentration. In addition, hyperglycemic effects have been observed in several acute OP poisoning in humans (Rahimi and Abdollahi 2007), which suggests that the induction of hyperglycemia is a possible mechanism through which OP exert their toxicological effects.

Chronic exposure to malathion (> 48 h) was associated with greater changes in blood glucose concentrations compared with acute exposure (< 48 h). It has been reported that the effects of malathion on blood glucose concentration is time-dependent, with the major changes in glucose occurring shortly after exposure (< 6 h) (Gupta 1974) because an increase in blood glucose is transitory (Lasram et al. 2008). Chronic exposure to malathion increased the frequency of hyperglycemia states; this could be induced by the cell death of β -pancreatic cells (Biarnés et al. 2002). Reduction in the

number of β -pancreatic cells has been associated with an exacerbation of hyperglycemia states (Anuradha et al. 2014).

The blood glucose could be metabolized through of glycolytic pathways to the finished product “pyruvate” that is metabolized into mitochondria by the pyruvate dehydrogenase (PDH) to acetyl coenzyme A (acetyl-CoA). The acetyl-CoA is condensed with oxaloacetate and it can then be oxidized; this process generates citrate (García-Ruiz et al. 2013). Citrate is a negative regulator for glycolysis enzymes (Usenik and Legiša 2010). The reduction of glycolysis activity disrupt the metabolism of glucose lead to the persistence of hyperglycemic; malathion has shown the capacity for decrease glycolytic enzymes (Ismail 2013), this could be a reason for the hyperglycemia.

In addition, the exposure to OP has been linked with the increase in de novo lipid synthesis by the overstimulation of fatty acid synthetase (FAS) (Howell et al. 2016). The effects of malathion on the activity of FAS are unknown. However, an increase in blood levels of triglycerides and cholesterol has been reported in animals exposed (Ismail 2013; Lasram et al. 2015a). The increase of fatty acid levels leads to the

Table 3 Alteration in some metabolic parameters in animal models exposed to malathion

Source	Dosage	Number	Duration	Outcomes and results (compared with controls)	Comments
Abdollahi et al. (2004)	0, 100, 200, and 400 ppm	6	4 weeks	↑Glu; ↑GP; ↑PEPCK; ↔BW	Hyperglycemic state was associated with the overstimulation of glycolysis and gluconeogenesis enzymes.
Chowdhury et al. (1980)	1 g/kg of BW	NA	48 h	↓Glucose uptake	The reduction of glucose uptake was observed without cellular damage in the intestine.
Gupta (1974)	50 mg/100 g of BW	NA	6 and 12 h	↑Glu; ↑GIY	The transitory hyperglycemic state was observed.
Ismail (2013)	1.28 mg/kg of BW	8	4 weeks	↓HK; ↓PK; ↓PFK; ↓LDH; ↓GPI; ↑Chol; ↑Tg, ↑AST; ↑ALP ↑ALT; ↑γGT	The co-application of vitamin C, decrease the hyperglycemic effects of malathion.
Johri et al. (2013)	NA	6	42 days	↑Glu; ↑AST; ↑ALT	The hyperglycemic state was observed.
Lasram et al. (2008)	400 mg/kg of BW	10	2–12 h	↑Glu; ↑GIY; ↑MDA	Reduction in pancreatic cellularity and the induction of oxidative stress were associated with a hyperglycemic state.
Lasram et al. (2014b)	200 mg/ kg of BW	12	28 days	↑Glu; ↓GIY; ↑AST; ↑ALT; ↑Ins; ↑HbA1c; ↑HOMA-IR; ↑MDA, ↓GSH; ↓SOD; ↑IL-1β	Hyperglycemic state was associated with the induction of insulin resistance, oxidative stress, and inflammatory response.
Lasram et al. (2015a)	200 mg/kg of BW	8	28 days	↑Glu; ↓GIY; ↑Ins; ↑HbA1c ↑HOMA-IR; ↑Chol; ↑Tg, ↓HDL-C; ↑LDL-C	The induction of dyslipidemia and insulin resistance lead to hyperglycemia state.
Nili-Ahmadabadi et al. (2013)	400 mg/kg of BW	6	2 h	↑OGGT; ↓Ins; ↑ROS; ↑MDA; ↑CPG; ↑8-OHdG	Acute exposure decreases the insulin secretion; with induction of oxidative stress in pancreatic cells.
Panahi et al. (2006)	100-, 200-, and 400-ppm or 3-, 15-, and 75 mg/kg of BW	6	4 weeks or 0.5 h	↑Glu; ↑GK; ↑GDH	Alterations in enzymes implicated in the insulin release were observed.
Ramu and Korner (1975)	2 g/kg of BW	8	6 h	↑Glu	The application of atropine inhibit the hyperglycemic effects of malathion.
Rezg et al. (2006)	100 mg/kg of BW	12	32 days	↑GLY; ↑HK	Glycogen levels increase without changes in blood glucose concentrations.
Rodrigues et al. (1986)	650 mg/kg of BW	NA	0.5–4 h	↑Glu	The transitory hyperglycemic state was observed.
Ruckmani et al. (2011)	NA	6	0.3–4 h or 7–28 days	↑Glu; ↓GSH; ↓SOD; ↓CAT; ↓GPx; ↓GR; ↓GSH	The induction of oxidative stress in pancreatic cells was associated with a hyperglycemic stage.
Tuzcu et al. (2014)	10 mg/kg of BW	NA	24 h	↑Glu; ↑Ins, ↑TOS; ↓TAC	Hepatic inflammatory damage was associated with the development of hyperglycemic state.
Wali et al. (1984)	5 mg/100 g of BW	6	45 days	↓Glucose uptake	The capacity for adsorbing glucose in the intestine was observed.

NA non-available data, BW body weight, ↑ increase, ↓ decrease, ↔ no change, *Glu* glucose, *GP* glycogen phosphorylase, *PEPCK* phosphoenolpyruvate carboxykinase, *GIY* glycogen, *HK* hexokinase, *PK* piruvatekinase, *PFK* phosphofructokinase, *LDH* lactate dehydrogenase, *GPI* glucose phosphate isomerase, *Chol* cholesterol, *Tg* triglyceride, *AST* aspartate aminotransferase, *ALT* alanine aminotransferase, *γGT* gamma glutamyl transferase, *ALP* alkaline phosphatase, *MDA* malondialdehyde, *Ins* insulin, *HbA1c* glycated hemoglobin, *HOMA-IR* homeostasis model assessment to assess insulin resistance, *GSH* reduced glutathione, *SOD* superoxide dismutase, *IL-1β* interleukin 1β, *HDL-C* high density lipoprotein cholesterol, *LDL-C* low density lipoprotein cholesterol, *GLYC* glycerol, *Lact* lactate, *GlyK* glycerol kinase, *GS3Kb* glycogen synthase kinase 3b, *OGGT* oral glucose tolerance test, *ROS* reactive oxygen species, *CPG* carbonyl groups, *8-OHdG* 8-hydroxy-2'-deoxyguanosine, *GK* glucokinase, *GDH* glutamate dehydrogenase, *HK* hexokinase, *GPx* glutathione peroxidase, *GR* glutathione reductase, *TOS* total oxidants, *TAC* total antioxidant capacity

activation of protein kinase C (PKC). PKC is a serine kinases leading to increase in serine phosphorylation in insulin receptor substrates (IRS) and a decrease in the activity de PI3K-Akt pathway; this results in the

reduction of glucose uptake stimulated by insulin lead to hyperglycemia (Mittra et al. 2008).

On another hand, the hyperglycemia observed in animals exposed to malathion has been linked with the oxidative stress

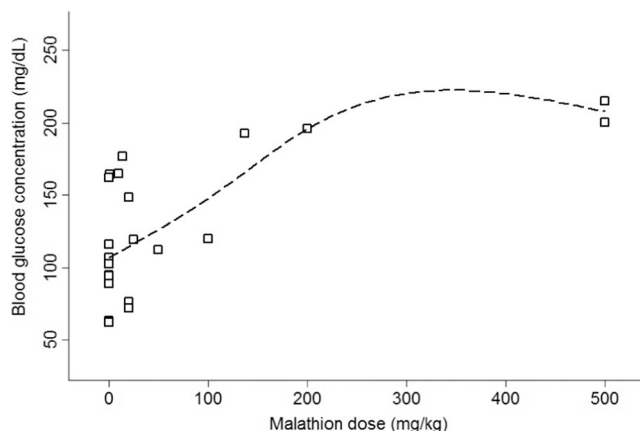


Fig. 3 Effect of malathion on the blood glucose concentration. Median spline plots from dose-response curves are presented. Squares indicate observed mean in individual studies. The dotted line is conformed of linear segments whose slopes represent the dose-response relationship in a specific range of values of the malathion dosage administered

induction (Lasram et al. 2008, 2014a, b; Ruckmani et al. 2011; Nili-Ahmadabadi et al. 2013; Tuzcu et al. 2014). This cellular condition is associated with the activation of the c-Jun N-terminal kinase (JNK). The JNK is a serine-kinase that increases the serine phosphorylation in IRS leads to their degradation with the negative regulation of PI3k-Akt pathway (Lee et al. 2003). This event reduces the GLUT4 translocation, resulting in a decrease in the glucose uptake and consumption that enhances the hyperglycemia (Tarantino and Caputi 2011).

The pooled analysis of the studies included in this work showed that exposure to malathion increased the blood glucose concentrations in exposed rats by 3.3-fold. The inter-individual variability in response to a toxin is a widely observed phenomenon; this variability has been attributed to genetic, phenotypic, and environmental factors and their interaction (Nebert 2005). In animal models, this variability has

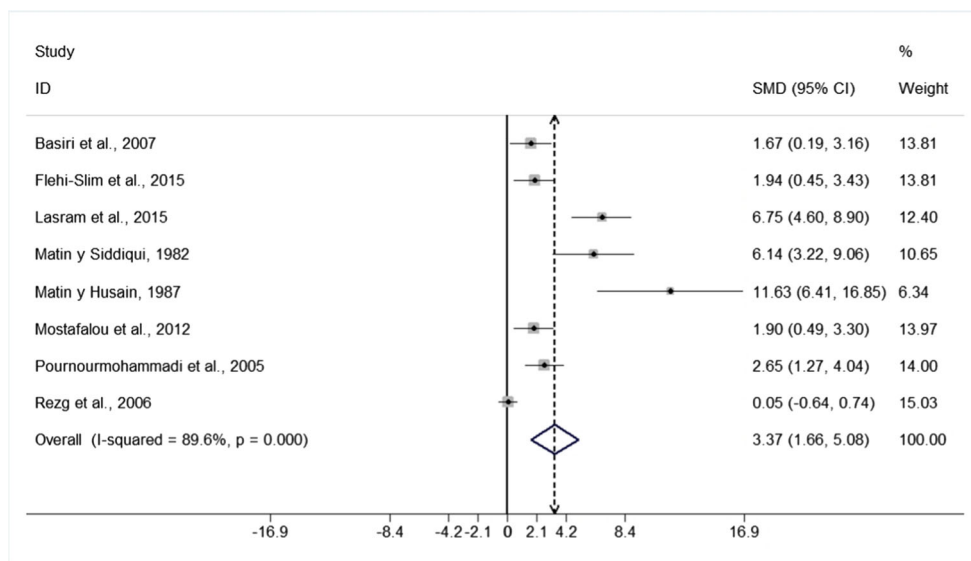
been observed. Figure 3 shows a high variability of basal (0 mg/kg) blood glucose concentrations, and inter-individual variability could be the cause of high and significant heterogeneity. In addition, meta-analyses of animal studies have often shown high heterogeneity attributed to differences in the species used and the design of the intervention protocols (Hooijmans et al. 2014a).

Data obtained in this work suggested that chronic or acute exposure to malathion increased the blood glucose concentration. However, only eight studies were included. The small number of articles used and the higher background inter-individual variability of the animals are some limitations. In addition, only two studies with a negative result were included; a possible overestimation of associations is feasible. However, this work agrees with a large body of evidence that shows a relation between OP exposure and hyperglycemia (Mostafalou 2016; Rahimi and Abdollahi 2007; Rezg et al. 2010; Mostafalou et al. 2012; Lasram et al. 2014a, b). The implementation of experimental models without high variability, such as in vitro models, is necessary to determine a possible mechanism of induced hyperglycemia after exposure to an OP.

Conclusions

In conclusion, the results obtained in this systematic review and meta-analysis of the primary studies based on animal models showed a positive/negative relationship between malathion exposure and the induction of a hyperglycemic state, validating the hyperglycemic effect of malathion. These results provide important information that consider the exposure to an OP as a possible risk factor for the development of insulin resistance.

Fig. 4 Forest plot of the blood glucose concentration after exposure to malathion generated from a random-effect model. The standardized mean difference (SMD) between the rats treated with malathion and control group with the 95% CIs is presented. The size of the gray box is proportional to the weight assigned to each study, and horizontal lines represent the 95% CIs



Acknowledgments The authors thank MBS. Teresa Domínguez Reyes for her technical assistance and data collection. This work was partially supported by Red Temática de Toxicología de Plaguicidas (CONACYT-253789/271775). RVMA was a recipient of a scholarship from CONACYT-Mexico (#276990).

References

- Abdollahi M, Donyavi M, Pournourmohammadi S, Saadat M (2004) Hyperglycemia associated with increased hepatic glycogen phosphorylase and phosphoenolpyruvate carboxykinase in rats following subchronic exposure to malathion. *Comp Biochem Physiol Part C Toxicol Pharmacol* 137(4):343–347. <https://doi.org/10.1016/j.cca.2004.03.009>
- Acker CI, Nogueira CW (2012) Chlorpyrifos acute exposure induces hyperglycemia and hyperlipidemia in rats. *Chemosphere* 89(5):602–608. <https://doi.org/10.1016/j.chemosphere.2012.05.059>
- Anuradha R, Saraswati M, Kumar KG, Rani SH (2014) Apoptosis of beta cells in diabetes mellitus. *DNA Cell Biol* 33(11):743–748. <https://doi.org/10.1089/dna.2014.2352>
- Arsenault AL, Gibson MA, Mader ME (1975) Hypoglycemia in malathion-treated chick embryos. *Can J Zool* 53(8):1055–1057. <https://doi.org/10.1139/z75-122>
- Barclay AW, Petocz P, McMillan-Price J, Flood VM, Prvan T, Mitchell P, Brand-Miller JC (2008) Glycemic index, glycemic load, and chronic disease risk—a meta-analysis of observational studies. *Am J Clin Nutr* 87(3):627–637
- Basiri S, Esmaily H, Vosough-Ghanbari S, Mohammadirad A, Yasa N, Abdollahi M (2007) Improvement by Satureja khuzestanica essential oil of malathion-induced red blood cells acetylcholinesterase inhibition and altered hepatic mitochondrial glycogen phosphorylase and phosphoenolpyruvate carboxykinase activities. *Pestic Biochem Physiol* 89(2):124–129. <https://doi.org/10.1016/j.pestbp.2007.04.006>
- Biamés M, Montolio M, Nacher V et al (2002) β -cell death and mass in syngeneically transplanted islets exposed to short- and long-term hyperglycemia. *Diabetes* 51(1):66–72. <https://doi.org/10.2337/diabetes.51.1.66>
- Bissell DM, Gores GJ, Laskin DL, Hoofnagle JH (2001) Drug-induced liver injury: mechanisms and test systems. *Hepatology* 33(4):1009–1013. <https://doi.org/10.1053/jhep.2001.23505>
- Brealey D, Singer M (2009) Hyperglycemia in critical illness: a review. *J Diabetes Sci Technol* 3(6):1250–1260. <https://doi.org/10.1177/193229680900300604>
- Chowdhury JS, Dudeja PK, Mehta SK, Mahmood A (1980) Effect of a single oral dose of malathion on D-glucose and glycine uptake and on brush border enzymes in rat intestine. *Toxicol Lett* 6(6):411–415. [https://doi.org/10.1016/0378-4274\(80\)90115-0](https://doi.org/10.1016/0378-4274(80)90115-0)
- Fakhri-Bafghi MS, Ghasemi-Niri SF, Mostafalou S et al (2016) Protective effect of selenium-based medicines on toxicity of three common organophosphorus compounds in human erythrocytes in vitro. *Cell J Yakhteh* 17:740–747
- Flehi-Slim I, Chargui I, Boughattas S, el Mabrouk A, Belaïd-Nouira Y, Neffati F, Najjar MF, Haouas Z, Cheikh HB (2015) Malathion-induced hepatotoxicity in male Wistar rats: biochemical and histopathological studies. *Environ Sci Pollut Res* 22(22):17828–17838. <https://doi.org/10.1007/s11356-015-5014-5>
- García-Ruiz C, Baulies A, Mari M, García-Rovés PM, Fernandez-Checa JC (2013) Mitochondrial dysfunction in non-alcoholic fatty liver disease and insulin resistance: cause or consequence? *Free Radic Res* 47(11):854–868. <https://doi.org/10.3109/10715762.2013.830717>
- Ghafoor-Rashidi Z, Dermenaki-Farahani E, Aliahmadi A, Esmaily H, Mohammadirad A, Ostad SN, Abdollahi M (2007) Protection by cAMP and cGMP phosphodiesterase inhibitors of diazinon-induced hyperglycemia and oxidative/nitrosative stress in rat Langerhans islets cells: molecular evidence for involvement of non-cholinergic mechanisms. *Pestic Biochem Physiol* 87(3):261–270. <https://doi.org/10.1016/j.pestbp.2006.08.007>
- Gould WW (1993) Linear splines and piecewise linear functions. *Stata Tech Bull* 15:13–17
- Gupta PK (1974) Malathion induced biochemical changes in rats. *Acta Pharmacol Toxicol (Copenh)* 35(3):191–194. <https://doi.org/10.1111/j.1600-0773.1974.tb00738.x>
- Harbord RM, Higgins JP (2008) Meta-regression in Stata. *Meta* 8:493–519
- Hooijmans CR, Int'Hout J, Ritskes-Hoitinga M, Rovers MM (2014a) Meta-analyses of animal studies: an introduction of a valuable instrument to further improve healthcare. *ILAR J* 55(3):418–426. <https://doi.org/10.1093/ilar/ilu042>
- Hooijmans CR, Rovers MM, de Vries RB et al (2014b) SYRCLE's risk of bias tool for animal studies. *BMC Med Res Methodol* 14(1):43. <https://doi.org/10.1186/1471-2288-14-43>
- Howell GE, Mulligan C, Young D, Kondakala S (2016) Exposure to chlorpyrifos increases neutral lipid accumulation with accompanying increased de novo lipogenesis and decreased triglyceride secretion in McArdle-RH7777 hepatoma cells. *Toxicol Vitro Int J Publ Assoc BIBRA* 32:181–189. <https://doi.org/10.1016/j.tiv.2016.01.002>
- Ismail SM (2013) Protective effects of vitamin C against biochemical toxicity induced by malathion pesticides in male albino rat. *J Evol Biol Res* 5:1–5
- Johri R, Dayal V, Johri PK (2013) In vivo effect of selected synthetic vegetable pesticides (malathion, chlorpyrifos and cypermethrin) with special reference to hematological and biochemical parameters of albino rat. *J Exp Zool India* 16:65–79
- Joshi AKR, Rajini PS (2009) Reversible hyperglycemia in rats following acute exposure to acephate, an organophosphorus insecticide: role of gluconeogenesis. *Toxicology* 257(1–2):40–45. <https://doi.org/10.1016/j.tox.2008.12.006>
- Joshi AKR, Rajini PS (2012) Hyperglycemic and stressogenic effects of monocrotophos in rats: evidence for the involvement of acetylcholinesterase inhibition. *Exp Toxicol Pathol* 64(1–2):115–120. <https://doi.org/10.1016/j.etp.2010.07.003>
- Karami-Mohajeri S, Abdollahi M (2011) Toxic influence of organophosphate, carbamate, and organochlorine pesticides on cellular metabolism of lipids, proteins, and carbohydrates: a systematic review. *Hum Exp Toxicol* 30(9):1119–1140. <https://doi.org/10.1177/0960327110388959>
- Karami-Mohajeri S, Hadian MR, Fouladdel S, Azizi E, Ghahramani MH, Hosseini R, Abdollahi M (2014) Mechanisms of muscular electrophysiological and mitochondrial dysfunction following exposure to malathion, an organophosphorus pesticide. *Hum Exp Toxicol* 33(3):251–263. <https://doi.org/10.1177/0960327113493300>
- Lasram MM, Annabi AB, Rezg R, Elj N, Slimen S, Kamoun A, El-Fazaa S, Gharbi N (2008) Effect of short-time malathion administration on glucose homeostasis in Wistar rat. *Pestic Biochem Physiol* 92(3):114–119. <https://doi.org/10.1016/j.pestbp.2008.06.006>
- Lasram MM, Dhoub IB, Annabi A, El Fazaa S, Gharbi N (2014a) A review on the molecular mechanisms involved in insulin resistance induced by organophosphorus pesticides. *Toxicology* 322:1–13. <https://doi.org/10.1016/j.tox.2014.04.009>
- Lasram MM, Dhoub IB, Bouzid K, Lamine AJ, Annabi A, Belhadjmida N, Ahmed MB, Fazaa SE, Abdelmoula J, Gharbi N (2014b) Association of inflammatory response and oxidative injury in the pathogenesis of liver steatosis and insulin resistance following subchronic exposure to malathion in rats. *Environ Toxicol*

- Pharmacol 38(2):542–553. <https://doi.org/10.1016/j.etap.2014.08.007>
- Lasram MM, Bouzid K, Douib IB, Annabi A, El Elj N, El Fazaa S, Abdelmoula J, Gharbi N (2015a) Lipid metabolism disturbances contribute to insulin resistance and decrease insulin sensitivity by malathion exposure in Wistar rat. *Drug Chem Toxicol* 38(2):227–234. <https://doi.org/10.3109/01480545.2014.933348>
- Lasram MM, El-Golli N, Lamine AJ, Douib IB, Bouzid K, Annabi A, El Fazaa S, Abdelmoula J, Gharbi N (2015b) Changes in glucose metabolism and reversion of genes expression in the liver of insulin-resistant rats exposed to malathion. The protective effects of N-acetylcysteine. *Gen Comp Endocrinol* 215:88–97. <https://doi.org/10.1016/j.ygcen.2014.10.002>
- Lassiter TL, Ryde IT, MacKillop EA, Brown KK, Levin ED, Seidler FJ, Slotkin TA (2008) Exposure of neonatal rats to parathion elicits sex-selective reprogramming of metabolism and alters the response to a high-fat diet in adulthood. *Environ Health Perspect* 116(11):1456–1462. <https://doi.org/10.1289/ehp.11673>
- Lee YH, Giraud J, Davis RJ, White MF (2003) c-Jun N-terminal kinase (JNK) mediates feedback inhibition of the insulin signaling cascade. *J Biol Chem* 278(5):2896–2902. <https://doi.org/10.1074/jbc.M208359200>
- Levitan EB, Song Y, Ford ES, Liu S (2004) Is nondiabetic hyperglycemia a risk factor for cardiovascular disease?: a meta-analysis of prospective studies. *Arch Intern Med* 164(19):2147–2155. <https://doi.org/10.1001/archinte.164.19.2147>
- Li L, Messina JL (2009) Acute insulin resistance following injury. *Trends Endocrinol Metab* TEM 20(9):429–435. <https://doi.org/10.1016/j.tem.2009.06.004>
- Liberati A, Altman DG, Tetzlaff J, et al (2009) The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *Ann Intern Med* 151:W–65
- Matin MA, Husain K (1987) Cerebral glycogenolysis and glycolysis in malathion-treated hyperglycaemic animals. *Biochem Pharmacol* 36(11):1815–1817. [https://doi.org/10.1016/0006-2952\(87\)90243-7](https://doi.org/10.1016/0006-2952(87)90243-7)
- Matin MA, Siddiqui RA (1982) Effect of diacetylmonoxime and atropine on malathion-induced changes in blood glucose level and glycogen content of certain brain structures of rats. *Biochem Pharmacol* 31(9):1801–1803. [https://doi.org/10.1016/0006-2952\(82\)90692-X](https://doi.org/10.1016/0006-2952(82)90692-X)
- Mittra S, Bansal VS, Bhatnagar PK (2008) From a glucocentric to a lipocentric approach towards metabolic syndrome. *Drug Discov Today* 13(5-6):211–218. <https://doi.org/10.1016/j.drudis.2008.01.006>
- Mostafalou S (2016) Persistent organic pollutants and concern over the link with insulin resistance related metabolic diseases. *Rev Environ Contam Toxicol* 238:69–89. https://doi.org/10.1007/398_2015_5001
- Mostafalou S, Abdollahi M (2016) Pesticides: an update of human exposure and toxicity. *Arch Toxicol* 91(2):1–51. <https://doi.org/10.1007/s00204-016-1849-x>
- Mostafalou S, Eghbal MA, Nili-Ahmadabadi A, Baeeri M, Abdollahi M (2012) Biochemical evidence on the potential role of organophosphates in hepatic glucose metabolism toward insulin resistance through inflammatory signaling and free radical pathways. *Toxicol Ind Health* 28(9):840–851. <https://doi.org/10.1177/0748233711425073>
- Namba T (1971) Cholinesterase inhibition by organophosphorus compounds and its clinical effects. *Bull World Health Organ* 44(1-3):289–307
- Nebert DW (2005) Inter-individual susceptibility to environmental toxicants—a current assessment. *Toxicol Appl Pharmacol* 207(2):34–42. <https://doi.org/10.1016/j.taap.2005.01.043>
- Nili-Ahmadabadi A, Pourkhalili N, Fouladdel S, Pakzad M, Mostafalou S, Hassani S, Baeeri M, Azizi E, Ostad SN, Hosseini R, Sharifzadeh M, Abdollahi M (2013) On the biochemical and molecular mechanisms by which malathion induces dysfunction in pancreatic islets in vivo and in vitro. *Pestic Biochem Physiol* 106(1-2):51–60. <https://doi.org/10.1016/j.pestbp.2013.04.003>
- Panahi P, Vosough-Ghanbari S, Pournourmohammadi S, Ostad SN, Nikfar S, Minaie B, Abdollahi M (2006) Stimulatory effects of malathion on the key enzymes activities of insulin secretion in Langerhans islets, glutamate dehydrogenase and glucokinase. *Toxicol Mech Methods* 16(4):161–167. <https://doi.org/10.1080/15376520500191623>
- Petersen KF, Shulman GI (2006) Etiology of insulin resistance. *Am J Med* 119(5):S10–S16. <https://doi.org/10.1016/j.amjmed.2006.01.009>
- Pournourmohammadi S, Farzami B, Ostad SN, Azizi E, Abdollahi M (2005) Effects of malathion subchronic exposure on rat skeletal muscle glucose metabolism. *Environ Toxicol Pharmacol* 19(1):191–196. <https://doi.org/10.1016/j.etap.2004.07.002>
- Raafat N, Abass MA, Salem HM (2012) Malathion exposure and insulin resistance among a group of farmers in Al-Sharkia governorate. *Clin Biochem* 45(18):1591–1595. <https://doi.org/10.1016/j.clinbiochem.2012.07.108>
- Rahimi R, Abdollahi M (2007) A review on the mechanisms involved in hyperglycemia induced by organophosphorus pesticides. *Pestic Biochem Physiol* 88(2):115–121. <https://doi.org/10.1016/j.pestbp.2006.10.003>
- Ramu A, Korner M (1975) Evidence of central influences on blood glucose level: malathion hyperglycemia. *Eur J Pharmacol* 32(1):120–123. [https://doi.org/10.1016/0014-2999\(75\)90331-3](https://doi.org/10.1016/0014-2999(75)90331-3)
- Rezg R, Mornagui B, El-Arbi M, Kamoun A, El-Fazaa S, Gharbi N (2006) Effect of subchronic exposure to malathion on glycogen phosphorylase and hexokinase activities in rat liver using native PAGE. *Toxicology* 223(1-2):9–14. <https://doi.org/10.1016/j.tox.2006.02.020>
- Rezg R, Mornagui B, Kamoun A, El-Fazaa S, Gharbi N (2007) Effect of subchronic exposure to malathion on metabolic parameters in the rat. *C R Biol* 330(2):143–147. <https://doi.org/10.1016/j.crv.2006.11.002>
- Rezg R, Mornagui B, El-Fazaa S, Gharbi N (2010) Organophosphorus pesticides as food chain contaminants and type 2 diabetes: a review. *Trends Food Sci Technol* 21(7):345–357. <https://doi.org/10.1016/j.tifs.2010.04.006>
- Rodrigues MA, Puga FR, Chenker E, Mazanti MT (1986) Short-term effect of malathion on rats' blood glucose and on glucose utilization by mammalian cells in vitro. *Ecotoxicol Environ Saf* 12(2):110–113. [https://doi.org/10.1016/0147-6513\(86\)90046-1](https://doi.org/10.1016/0147-6513(86)90046-1)
- Romero-Navarro G, Lopez-Aceves T, Rojas-Ochoa A, Mejia CF (2006) Effect of dichlorvos on hepatic and pancreatic glucokinase activity and gene expression, and on insulin mRNA levels. *Life Sci* 78(9):1015–1020. <https://doi.org/10.1016/j.lfs.2005.06.010>
- Ruckmani A, Nayar PG, Konda VGR, Madhusudha N, Madhavi E, Chokkaling M, Meti V, Sundaraval S (2011) Effects of inhalational exposure of malathion on blood glucose and antioxidants level in Wistar albino rats. *Res J Environ Toxicol* 5(5):309–315. <https://doi.org/10.3923/rjet.2011.309.315>
- Sodhi S, Sharma A, Brar APS, Brar RS (2008) Effect of α tocopherol and selenium on antioxidant status, lipid peroxidation and hepatopathy induced by malathion in chicks. *Pestic Biochem Physiol* 90(2):82–86. <https://doi.org/10.1016/j.pestbp.2007.08.002>
- Stead LG, Gilmore RM, Bellolio MF, Mishra S, Bhagra A, Vaidyanathan L, Decker WW, Brown RD Jr (2009) Hyperglycemia as an independent predictor of worse outcome in non-diabetic patients presenting with acute ischemic stroke. *Neurocrit Care* 10(2):181–186. <https://doi.org/10.1007/s12028-008-9080-0>
- Sterne JA, Bradburn MJ, Egger M (2008) Meta-analysis in Stata™. *Syst Rev Health Care Meta-Anal Context Second Ed* 347–369
- Tarantino G, Caputi A (2011) JNKs, insulin resistance and inflammation: a possible link between NAFLD and coronary artery disease. *World*

- J Gastroenterol 17(33):3785–3794. <https://doi.org/10.3748/wjg.v17.i33.3785>
- Teimouri F, Amirkabirian N, Esmaily H, Mohammadirad A, Aliahmadi A, Abdollahi M (2006) Alteration of hepatic cells glucose metabolism as a non-cholinergic detoxication mechanism in counteracting diazinon-induced oxidative stress. *Hum Exp Toxicol* 25(12):697–703. <https://doi.org/10.1177/0960327106075064>
- Tomás E, Lin Y-S, Dagher Z et al (2002) Hyperglycemia and insulin resistance: possible mechanisms. *Ann N Y Acad Sci* 967(1):43–51. <https://doi.org/10.1111/j.1749-6632.2002.tb04262.x>
- Tuzcu K, Alp H, Ozgur T, Karcioğlu M, Davarci I, Evliyaoglu O, Karakus A, Hakimoglu S (2014) Oral intralipid emulsion use: a novel therapeutic approach to pancreatic β -cell injury caused by malathion toxicity in rats. *Drug Chem Toxicol* 37(3):261–267. <https://doi.org/10.3109/01480545.2013.838780>
- Usenik A, Legiša M (2010) Evolution of allosteric citrate binding sites on 6-phosphofructo-1-kinase. *PLoS One* 5(11):e15447. <https://doi.org/10.1371/journal.pone.0015447>
- Vosough-Ghanbari S, Sayyar P, Pournourmohammadi S, Aliahmadi A, Ostad SN, Abdollahi M (2007) Stimulation of insulin and glucagon synthesis in rat Langerhans islets by malathion in vitro: evidence for mitochondrial interaction and involvement of subcellular non-cholinergic mechanisms. *Pestic Biochem Physiol* 89(2):130–136. <https://doi.org/10.1016/j.pestbp.2007.05.001>
- Wali RK, Singh R, Dudeja PK, Sarkar AK, Mahmood A (1984) Subchronic malathion treatment effects on rat intestinal functions. *Bull Environ Contam Toxicol* 33(1):289–294. <https://doi.org/10.1007/BF01625545>
- World Health Organization (WHO) (2010) The WHO recommended classification of pesticides by hazard and guidelines to classification 2009. World Health Organization, Geneva